

ECE269: Linear Algebra and Applications
Winter 2026
Discussion 6

In lectures, we have proved that any subspace $\mathcal{V}$ with a finite spanning set has a basis $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$. In this discussion we prove by construction that:

Theorem 1. *If a subspace $\mathcal{V}$ is endowed with an inner product $\langle \cdot, \cdot \rangle$, and it has a basis, then it has an orthonormal basis.*

To prove this we begin with proving that the **Gram-Schmidt Orthogonalization** method produces a basis $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ such that:

1. $\text{span}(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n) = \text{span}(\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$.
2. $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ is a set of orthonormal vectors, i.e.,

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}.$$

1 Gram-Schmidt Procedure

Let $\|\cdot\|_{\text{ip}}$ be the inner product induced norm, then the Gram-Schmidt procedure constructs this basis by iterative projection and normalization as follows:

$$\begin{aligned} \tilde{\mathbf{u}}_1 &\leftarrow \mathbf{b}_1 \\ \mathbf{u}_1 &\leftarrow \frac{\tilde{\mathbf{u}}_1}{\|\tilde{\mathbf{u}}_1\|_{\text{ip}}} \\ \tilde{\mathbf{u}}_2 &\leftarrow \mathbf{b}_2 - \langle \mathbf{b}_2, \mathbf{u}_1 \rangle \mathbf{u}_1 \\ \mathbf{u}_2 &\leftarrow \frac{\tilde{\mathbf{u}}_2}{\|\tilde{\mathbf{u}}_2\|_{\text{ip}}} \\ \tilde{\mathbf{u}}_3 &\leftarrow \mathbf{b}_3 - \langle \mathbf{b}_3, \mathbf{u}_1 \rangle \mathbf{u}_1 - \langle \mathbf{b}_3, \mathbf{u}_2 \rangle \mathbf{u}_2 \\ \mathbf{u}_3 &\leftarrow \frac{\tilde{\mathbf{u}}_3}{\|\tilde{\mathbf{u}}_3\|_{\text{ip}}} \\ &\vdots \\ \tilde{\mathbf{u}}_n &\leftarrow \mathbf{b}_n - \langle \mathbf{b}_n, \mathbf{u}_1 \rangle \mathbf{u}_1 - \langle \mathbf{b}_n, \mathbf{u}_2 \rangle \mathbf{u}_2 - \dots - \langle \mathbf{b}_n, \mathbf{u}_{n-1} \rangle \mathbf{u}_{n-1} \\ \mathbf{u}_n &\leftarrow \frac{\tilde{\mathbf{u}}_n}{\|\tilde{\mathbf{u}}_n\|_{\text{ip}}} \end{aligned}.$$

We now prove that $\text{span}(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n) = \text{span}(\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$, and $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ is a set of orthonormal vectors.

Proof. The Gram-Schmidt basis are given by

$$\mathbf{u}_1 = \frac{\mathbf{b}_1}{\|\mathbf{b}_1\|_{\text{ip}}}, \quad \mathbf{u}_k = \frac{\mathbf{b}_k - \sum_{i=1}^{k-1} \langle \mathbf{b}_k, \mathbf{u}_i \rangle \mathbf{u}_i}{\|\mathbf{b}_k - \sum_{i=1}^{k-1} \langle \mathbf{b}_k, \mathbf{u}_i \rangle \mathbf{u}_i\|_{\text{ip}}}, \quad n \geq k \geq 2. \quad (1)$$

We use proof by induction. First see that $\text{span}(\mathbf{b}_1) = \text{span}(\mathbf{u}_1)$ since $\mathbf{u}_1$ is a non-zero multiple of $\mathbf{b}_1$. Moreover, $\|\mathbf{u}_1\|_{\text{ip}} = 1$. Now for the induction step, we will assume that

$$\text{span}(\mathbf{b}_1, \dots, \mathbf{b}_{j-1}) = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}), \quad n \geq j \geq 2,$$

where $\{\mathbf{u}_1, \dots, \mathbf{u}_{j-1}\}$ are orthonormal vectors. Note that $\mathbf{b}_j \notin \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1})$, because the vectors $\{\mathbf{b}_1, \dots, \mathbf{b}_j\}$ are linearly independent. Hence, the vector

$$\mathbf{u}_j = \frac{\mathbf{b}_j - \sum_{i=1}^{j-1} \langle \mathbf{b}_j, \mathbf{u}_i \rangle \mathbf{u}_i}{\|\mathbf{b}_j - \sum_{i=1}^{j-1} \langle \mathbf{b}_j, \mathbf{u}_i \rangle \mathbf{u}_i\|_{\text{ip}}}$$

is well-defined and $\|\mathbf{u}_j\|_{\text{ip}} = 1$.

Now, for any $1 \leq k \leq j-1$

$$\langle \mathbf{u}_j, \mathbf{u}_k \rangle = \frac{\langle \mathbf{b}_j, \mathbf{u}_k \rangle - \sum_{i=1}^{j-1} \langle \mathbf{b}_j, \mathbf{u}_i \rangle \langle \mathbf{u}_i, \mathbf{u}_k \rangle}{\|\mathbf{b}_j - \sum_{i=1}^{j-1} \langle \mathbf{b}_j, \mathbf{u}_i \rangle \mathbf{u}_i\|_{\text{ip}}},$$

where we used the linearity in the first argument of the inner product. But $\langle \mathbf{u}_i, \mathbf{u}_k \rangle = \delta_{ik}$ due to our initial assumption. Hence, we have

$$\langle \mathbf{u}_j, \mathbf{u}_k \rangle = \frac{\langle \mathbf{b}_j, \mathbf{u}_k \rangle - \langle \mathbf{b}_j, \mathbf{u}_k \rangle}{\|\mathbf{b}_j - \sum_{i=1}^{j-1} \langle \mathbf{b}_j, \mathbf{u}_i \rangle \mathbf{u}_i\|_{\text{ip}}} = 0.$$

Hence $\{\mathbf{u}_1, \dots, \mathbf{u}_{j-1}, \mathbf{u}_j\}$ are orthonormal vectors.

By our induction assumption $\mathbf{b}_k \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}, \mathbf{u}_j)$ for $1 \leq k \leq j-1$. Since $\mathbf{b}_j = \alpha \mathbf{u}_j + \sum_{i=1}^{j-1} \langle \mathbf{b}_j, \mathbf{u}_i \rangle \mathbf{u}_i$, where $\alpha = \|\mathbf{b}_j - \sum_{i=1}^{j-1} \langle \mathbf{b}_j, \mathbf{u}_i \rangle \mathbf{u}_i\|_{\text{ip}}$, then $\mathbf{b}_j \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}, \mathbf{u}_j)$. This leads to

$$\text{span}(\mathbf{b}_1, \dots, \mathbf{b}_{j-1}, \mathbf{b}_j) \subseteq \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}, \mathbf{u}_j).$$

Now $\dim(\text{span}(\mathbf{b}_1, \dots, \mathbf{b}_{j-1}, \mathbf{b}_j)) = j$, since $\{\mathbf{b}_1, \dots, \mathbf{b}_{j-1}, \mathbf{b}_j\}$ are linearly independent. Similarly, $\dim(\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}, \mathbf{u}_j)) = j$, since $\{\mathbf{u}_1, \dots, \mathbf{u}_{j-1}, \mathbf{u}_j\}$ are orthogonal and hence linearly independent. Therefore $\text{span}(\mathbf{b}_1, \dots, \mathbf{b}_{j-1}, \mathbf{b}_j) = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}, \mathbf{u}_j)$.

Since both the base step and induction step hold true, by induction, $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ is an orthonormal basis for $\text{span}(\mathbf{b}_1, \dots, \mathbf{b}_n)$. This exercise shows us that every finite-dimensional subspace has an orthonormal basis provided that underlying vector space is equipped with an inner product, and the Gram-Schmidt procedure gives us a way to construct such an orthonormal basis from any given basis. $\square$

2 QR Decomposition

Suppose $\mathcal{S}$ is k -dimensional subspace of $\mathbb{R}^n$. Let $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k\}$ be a basis for $\mathcal{S}$. Using (1), we can write $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_j\}$ in terms of $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_j\}$, we have:

$$\begin{aligned}\mathbf{b}_j &= \langle \mathbf{b}_j, \mathbf{u}_1 \rangle \mathbf{u}_1 + \langle \mathbf{b}_j, \mathbf{u}_2 \rangle \mathbf{u}_2 + \dots + \langle \mathbf{b}_j, \mathbf{u}_{j-1} \rangle \mathbf{u}_{j-1} + \|\tilde{\mathbf{u}}_j\|_{\text{ip}} \mathbf{u}_j \\ &= r_{j1} \mathbf{u}_1 + r_{j2} \mathbf{u}_2 + \dots + r_{jj-1} \mathbf{u}_{j-1} + r_{jj} \mathbf{u}_j\end{aligned},$$

for $j = 1, 2, \dots, k$. This can be rewritten in the matrix form as

$$\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \dots, \mathbf{b}_k] = \underbrace{[\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_k]}_{\mathbf{Q}} \underbrace{\begin{bmatrix} r_{11} & r_{21} & \dots & r_{k1} \\ 0 & r_{22} & \dots & r_{k2} \\ 0 & 0 & \dots & r_{k3} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & r_{kk} \end{bmatrix}}_{\mathbf{R}} = \mathbf{QR} \quad .$$

where $\mathbf{Q} \in \mathbb{R}^{n \times k}$ consists of orthonormal columns (i.e. $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_{k \times k}$), and $\mathbf{R} \in \mathbb{R}^{k \times k}$ is upper triangular with nonzero diagonal entries. This factorization of the matrix $\mathbf{B}$ into an orthogonal matrix and an upper triangular matrix is often referred to as a **QR**-decomposition of $\mathbf{B}$.

The same procedure can be easily extended to an arbitrary set of vectors by slightly modifying the Gram–Schmidt procedure. Suppose that $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \dots, \mathbf{b}_k\}$ are dependent, then some $\mathbf{b}_j$ is a linear combination of $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3, \dots, \mathbf{b}_{j-1}\}$. The corresponding $\tilde{\mathbf{u}}_j$ obtained in (1) would be zero and thus cannot be normalized. In the modified procedure, we simply skip such zero vectors, projecting each $\mathbf{b}_j$ into proper orthonormal vectors obtained thus far. This simple modification results in the following factorization

$$\mathbf{B} = \underbrace{[\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_q]}_{\mathbf{Q}} \underbrace{\begin{bmatrix} r_{11} & r_{21} & \dots & r_{k1} \\ 0 & r_{22} & \dots & r_{k2} \\ 0 & 0 & \dots & r_{k3} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & r_{kq} \end{bmatrix}}_{\mathbf{R}} = \mathbf{QR} \quad ,$$

where $\mathbf{Q} \in \mathbb{R}^{n \times q}$ consists of $q = \text{rank}(\mathbf{B})$ orthonormal columns, and $\mathbf{R} \in \mathbb{R}^{q \times k}$ is in the upper staircase form. Note that in general QR-decomposition is not unique. **QR**-decomposition renders solving linear equations much easier.